B. Rajesh

Reg No: 192526022-6

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : _____ Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing $P(X = 2 \mid Y = 1)$, (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification. **(6) (CO1) (BL4)**

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.) **(4) (CO1) (BL4)**

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality. **(3) (CO1) (BL4)**

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P‖Q) and KL(Q‖P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss. **(3) (CO1) (BL4)**

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A. **(4) (CO1) (BL4)**

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases. **(4) (CO1) (BL4)**

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal. **(3) (CO1) (BL5)**

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: _____

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	4
Task 4	4	3
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	3
Task 9	3	3
Task 10	4	4
Task 11	4	3
Task 12	8	8
TOTAL	**50**	42

Total in words:

Assignment - 1

Unit - 1

B. Rajesh
RgNo: 19252b0026

Code: MLA0L

Name: Fundamentals of Machine learning

1.A) Design of learning system

Machine learning system are designed using mitchell's three components

Case	learning type	learning task
(a) loan default prediction	Supervised	classification
(b) Customer segmentation	unsupervised	clustering
(c) Credit Card spending	supervised	Regression
(d) ATM cash refill	Reinforcement	control
(e) Fraud detection	semi-supervised	classification

(A) Loan default prediction: - A system predict whether a customer will repay a loan

(B) customer segmention: - The system groups coustomers according to similar purchasing behaviour

(C) Credit card Spending: - The system predict the next mounth spending amount in rupesses

(D) ATM cash Refill: - The system learns the best cash refill policy

(E) Fraud Detection: The system identifies fraudulent transisactions using 2% labelled 98% unlabelled data

-5-

2A) **Probability of Disease Detection**

Given

Disease prevalence = 2%

sensitivity = 95%

specificity = 90%

using Bayes' theorem

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)}$$

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.117}$$

$$P(D|+) = 16.24\%$$

3A) **Joint probability Distribution:-**

Given

x	y=0	y=1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

(a) Marginal Probabilities:

for x :-

$$P(x=0) = 0.30$$
$$P(x=1) = 0.40$$
$$P(x=2) = 0.30$$

for y :-

$$P(y=0) = 0.45$$
$$P(y=1) = 0.55$$

(b) expected values & variance

$$E(x) = \Sigma x P(x)$$
$$= 0(0.30) + 1(0.40) + 2(0.30)$$
$$\boxed{E(x) = 1}$$
$$E(x^2) = 0 + 0.40 + 1.20 = 1.60$$
$$\text{var}(x) = E(x^2) - (E(x))^2$$
$$= 1.60 - 1 = 0.60$$

-4-

$$E(y) = 0(0.45) + 1(0.55)$$
$$\boxed{E(y) = 0.55}$$

(c) conditional Probability

$$P(x=2 \mid y=1) = \frac{P(x=2, y=1)}{P(y=1)}$$
$$= \frac{0.10}{0.55}$$
$$\boxed{0.1818}$$

4.A) Normal Distribution and maximum likelihood estimation

(a) calculate Probability

$$x \sim N(50, 16)$$

$$\mu = 50, \sigma = 4$$

formula:

$$z = \frac{x - \mu}{\sigma}$$

for $P(x > 58)$

$$z = \frac{58 - 50}{4} = 2$$

$$P(x > 58) = 1 - 0.9772$$

0.0228

(b) maximum likihood estimation

Given reading 12, 15, 11, 14, 13

$$\mu = \frac{12 + 15 + 11 + 14 + 13}{5} = 13$$

$$\sigma^2_{ML} = \frac{\Sigma (x_i - \mu)^2}{n}$$

$$\frac{1 + 4 + 4 + 1 + 0}{5} = 2$$

variance

$$\therefore s^2 = \frac{10}{5 - 1}$$

$= 2.5$

Entropy & Average code length

Given

$$P = (1/2, 1/4, 1/8, 1/8)$$

(a) Entropy

$$H(x) = -\sum P(x) \log_2 P(x)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$\boxed{H(x) = 1.75 \text{ bits}}$$

(b) uniform source entropy

$$H(x) = \log_2 4$$

$$\boxed{H(x) = 2 \text{ bits}}$$

Redundancy $R = 2 - 1.75 = \boxed{0.25 \text{ bits}}$

3

Percentage $= \frac{0.25}{2} \times 100$

$= 12.5\ \%$

(c) Average code length

code = 0, 10, 110, 111

Their lengths 1, 2, 3, 3

$$L = 1/2(1) + 1/4(2) + 1/8(3) + 1/8(3)$$

$$\boxed{L = 1.75 \text{ bits}}$$

Q1. KL Divergence

Given

$$P = (0.5, 0.25, 0.25)$$

$$Q = (1/3, 1/3, 1/3)$$

(a) KL (P||Q)

$$D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$$

$$= 0.5 \log_2(1.5) + 0.5 \log_2(0.75)$$

$$D_{KL}(P||Q) = 0.085 \text{ bits}$$

(b) KL (Q||P)

$$D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$= 1/3 \log_2 2/3 + 2/3 \log_2 4/3$$

$$= D_{KL}(Q||P) = 0.0817 \text{ bits}$$

* KL divergence measure the difference b/w two probability distributions. It is not symmetric, meaning reversing P & Q changes the results

(a) Entropy and Mutual Information.

Given $P(0,0) = 0.4$, $P(0,1) = 0.1$

$P(1,0) = 0.1$, $P(1,1) = 0.4$

Entropy of A :-

$P(A=0) = 0.5$, $P(A=1) = 0.5$

$$H(A) = -\sum P(A) \log_2 P(A)$$

$$H(A) = 1 \text{ bit}$$

(b) Entropy of B :-

$$H(B) = 1 \text{ bit}$$

(c) Joint entropy $H(A,B) = -\sum P(a,b) \log_2 P(a,b)$

$$= -2(0.4 \log_2 0.4) - 2(0.1 \log_2 0.1)$$

$$H(A,B) = 1.722 \text{ bits}$$

(d) Mutual information :-

$$I(A,B) = H(A) - H(A/B)$$

$$= 1 - 0.722$$

$$= I(A;B) = 0.278 \text{ Bits}$$

8A) Bayes Decision Theory

Given $P(x|\omega_1) = N(2,4)$

$P(x|\omega_2) = N(6,4)$

$P(\omega_1) = 0.6, \quad P(\omega_2) = 0.4.$

(a) Decision threshold

For equal variances & zero one loss:

$$x^* = \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2}{\mu_2-\mu_1} \ln \frac{P(\omega_1)}{P(\omega_2)}$$

$$= 4 + \ln(1.5)$$

$$x^* = 4.4055$$

(b) Classify $x = 4.5$

since:

$$4.5 > 4.4055$$

$P(\text{error}) \simeq 0.154$

(c) Effect of increasing Prior probability.

When the prior probability of class ω_1 increase, the decision threshold moves towards the right. This increase the region classified as class ω_1

a11) Minimum Risk Decision

Given $P(w_1/x) = 0.3$, $P(w_2/x) = 0.7$

(a) Condition risk

$$R(\alpha_1/x) = 0.(0.3) + 8(0.7)$$

$$= 5.6$$

$$R(\alpha_2/x) = 50(0.3) + 0(0.7)$$

$$= 15$$

(b) minimum risk action

since 5.6 is less than 15 :-

α_1 = decleas disease

(c) Posterior probability threshold

Let $P = P(w_1/x)$

$$8(1-P) < 50P$$

$$8 < 58P$$

$$P > 0.1379 = 13.79\%$$

10.a) Bias - Variance Trade - off.

Degree	Training MSE	validation MSE
1	9.0	9.3
2	4.1	4.6
3	2.2	2.9
5	0.9	3.6
9	0.1	11.8.

(a) Generalisation gap

Degree	Generalisation gap
1	0.3
2	0.5
3	0.7
5	2.7
9	11.7.

4

(b) under fitting and over fitting

1 under fitting because the model is too simple
2 good generalisation
3 Best validation performance
4,5. shows overfitting
a. severe over fitting because training error is very low

(c) Two techniques to improve degree g-model

1. Regularisation
2. Reduce Polynomial degree.

n) Maximum likelihood & map estimation

Given

Total coins = 10

Heads = 7

Tails = 3

Prior distribution = Beta (2,4)

(a) maximum

$$\hat{\theta}_{ML} = \frac{\text{Heads}}{\text{Total tosses}}$$

$$= 7/10$$

$$\hat{\theta}_{ML} = 0.7$$

(b) maximum (map

$$\hat{\theta}_{MAP} = \frac{\alpha + H - 1}{\alpha + \beta + H + T - 2}$$

$$= \frac{2+7-1}{2+2+10-2}$$

$$= 8/12$$

(c) If 70 heads were $\hat{\theta}_{MAP} = 0.6667$ in 100 tosses

Estimate	Calculation	Result
MLE	70/100	0.7000
MAP	71/102	0.6961
Posterior mean	72/104	0.6923

11.A) Sepsis - Early warning system

(a)

(1.) Task(T) ÷ predict the possibility of sepsis or not

(2) Performance ÷ Measure the model using sensitivity

(3) Experience: using historical electronic health records containing vital signs

(4) Target Function: Predict whether a patient will develop sepsis or not

(5) Learning Algorithm: A supervised classification algorithm such as Random forest can be used

(6) validation Strategy: Divide the dataset into training validation and testing sets. to evaluate the model

(b) calculate TP, FN, FP, & TN.

Given

Total patient = 1000

Prevalence = 5%

Sensitivity = 90%

Specificity = 80%

Sepsis patients $1000 \times 0.05 = 50$

healthy patient $1000 - 50 = 950$

True Positive

$TP = 50 \times 0.90 = 45$

False Negative

$$FN = 50.45 = 3$$

True Negative

$$TN = 951 \times 0.80 = 760$$

False Positive

$$FP = 950 - 760 = 190$$

(c) Deployment Policy

The hospital use the machine learning model with human medical supervision. Doctors should review the alerts and make the final decision

$$100p > 5(1-p)$$

$$105p > 5$$

$$p > 0.0476 = 4.76\%$$